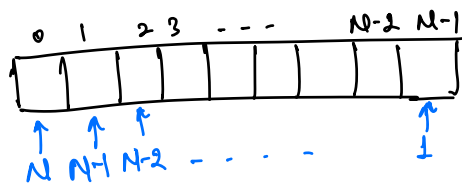


- Revise Subarray
- Subsequences vs subsets
- Check subset with given sum → Advance Question
- Sum of all subsets. → Advance Question
- Sum of max of all subsets. → Very Advance Question

Subarray Basics

Subarray:- continuous part of an array

 → Given N elements, how many subarrays?



$$\text{Total subarrays} = \frac{N(N+1)}{2}$$

Subsequences: Sequence generated by deleting 0 or more elements from your array.

arr[8] :-

	0	1	2	3	4	5	6	7
3	✓	✓	✓	✓	✓	✓	✓	✓
-2	✓	✓	✓	✓	✓	✓	✓	✓
0	✓	✓	✓	✓	✓	✓	✓	✓
1	✓	✓	✓	✓	✓	✓	✓	✓
8	✓	✓	✓	✓	✓	✓	✓	✓
7	✓	✓	✓	✓	✓	✓	✓	✓
4	✓	✓	✓	✓	✓	✓	✓	✓
9	✓	✓	✓	✓	✓	✓	✓	✓

→ Deleting 0 elements from the array.

- ① { -2, 0, 8, 4 } → Yes, it is a subsequence
- ② { 3, -2, 0, 1 } → Yes
- ③ { -2, 8, 7, 9 } → Yes
- ④ { } → Yes, deleting all elements from the given arr[]
- ⑤ { 1, 0, 8, 4, 9 } → NO
 { 0, 1, 8, 4, 9 } → Not same

arr[8] :-

	0	1	2	3	4	5	6	7
3	✓	✓	✓	✓	✓	✓	✓	✓
-2	✓	✓	✓	✓	✓	✓	✓	✓
0	✓	✓	✓	✓	✓	✓	✓	✓
1	✓	✓	✓	✓	✓	✓	✓	✓
8	✓	✓	✓	✓	✓	✓	✓	✓
7	✓	✓	✓	✓	✓	✓	✓	✓
4	✓	✓	✓	✓	✓	✓	✓	✓
9	✓	✓	✓	✓	✓	✓	✓	✓

→ { 0, 1, 8, 4, 9 }

In a subsequence, ordering/indexing is maintained

Quiz Not a subsequence :-

arr1 = {1, 2, 3, 4, 5}

a) {1, 2, 3, 4, 5} ✓

b) {4} ✓

c) {2, 3, 5} ✓

d) {5, 4, 3} ✗

Note:- Subsequences are ordered based on index values.

Subarrays vs Subsequences

arr[5] = ⁰{-3, ¹0, ²1, ³2, ⁴6}

	Subarray	Subsequence
{1 2 6}	✓	✓
{-3 1 2}	X	✓
{0 1 2}	✓	✓
{-3 1 6}	X	✓
{6 1 0}	X	X

→ All subarrays are subsequences.
→ But all subsequences are NOT subarrays

Sorting in a Subsequence

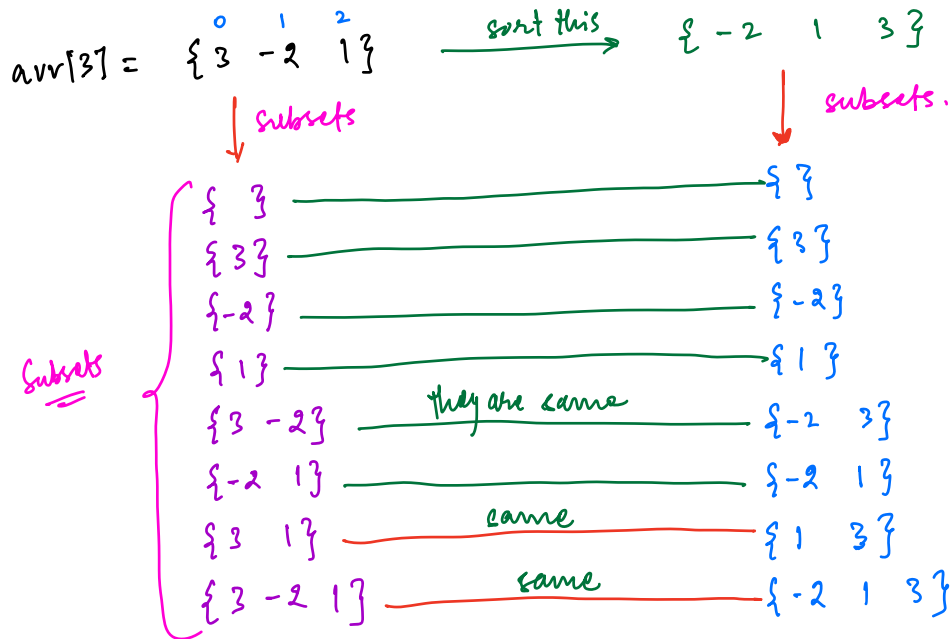
arr[3] = ⁰{3, ¹-2, ²1} $\xrightarrow{\text{sort this}}$ {-2 1 3}

↓ subsequences ↓ subsequences

{3}	→	{}
{3}	→	{3}
{-2}	→	{-2}
{1}	→	{1}
{3 -2}	→ NOT SAME	{-2 3}
{-2 1}	→	{-2 1}
{3 1}	→ not same	{1 3}
{3 -2 1}	→ not same	{-2 1 3}

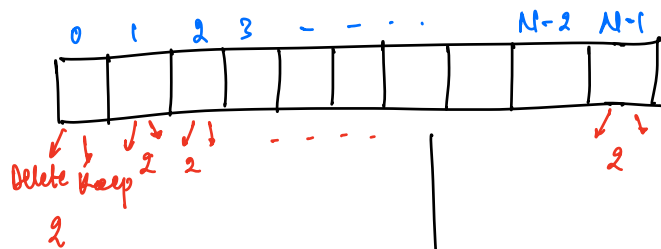
Note:- If you sort your i/p array your subsequences will change and you will not get correct ans.

Subsets :- Exactly same as a subsequence, but here the order won't matter



Can you sort data for subsets? → Yes.

Q1 Given N elements. Count number of subsequences.

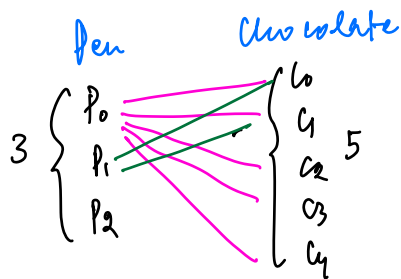


$= 2^N$ subsequences

$\{ \}$
empty

$\{ \text{entire array} \}$

Idea:- Valentine's Day



No. of subsets $= 2^N$

total choices $= 3 \times 5 = 15$

* We cannot have duplicate elements in a set
* A subset will only have unique elements

e.g. $\{ \overset{0}{1} \overset{1}{2} \overset{2}{2} \}$

Subsequences:- $\{ \}, \{1\}, \{2\}, \{2\}, \{1, 2\}, \{2, 2\}, \{1, 2, 2\}$

$\Rightarrow 2^N$ subsequences.

e.g. $\{ \overset{0}{1} \overset{1}{2} \overset{2}{2} \}$

Subsets :- $\{ \}, \{1\}, \{2\}, \{2\}, \{1, 2\}, \{2, 2\}, \{1, 2, 2\}$

counted as 1.
Because order does not matter in subsets.

$\{1, 2, 2\}$ is counted as 1.

Subsets $\Rightarrow 4$

$\{ \}, \{1\}, \{2\}, \{1, 2\}$

Note:- If data is repeating in i/p arr¹, subsets won't be 2^N

Given $N \rightarrow 2^N$ subsequences

Given N distinct elements $\rightarrow 2^N$ subsets

Q2 Given N distinct elements
 Check if there exists a subset with $\text{sum} = k$ $\rightarrow$ given as i/p.
 boolean

eg. 1. $\text{arr}[7] = \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & -1 & 0 & 6 & 2 & -3 & 5 \end{matrix}$
 $k = 10 \quad \left. \begin{matrix} \{3, 2, 5\} \\ \{6, 5, -1\} \\ \{3, -1, 6, 2\} \end{matrix} \right\} \rightarrow \text{return True}$

$k = 20 \rightarrow \text{return False}$

Idea:-

For every subset, get sum and then compare with k .

eg. $\text{arr}[3] = \{ \overset{0}{3}, \overset{1}{-2}, \overset{2}{1} \}$

No. of subsets $= 2^3 = 8$

	2	1	0	
0:	0	0	0	$\rightarrow \{ \}$
1:	0	0	1	$\rightarrow \{ 3 \}$
2:	0	1	0	$\rightarrow \{ -2 \}$
3:	0	1	1	$\rightarrow \{ 3, -2 \}$
4:	1	0	0	$\rightarrow \{ 1 \}$
5:	1	0	1	$\rightarrow \{ 3, 1 \}$
6:	1	1	0	$\rightarrow \{ -2, 1 \}$
7:	1	1	1	$\rightarrow \{ 3, -2, 1 \}$

3 elements
 $\rightarrow$ Hence, 8 subsets. $[0, 7]$

N elements
 $\rightarrow 2^N$ subsets, $[0, 2^N - 1]$

Pseudo Code:-

```
for (int i=0; i < 2N; i++) {  
    // for i, get all set bit positions  
    sum = 0;  
    for (int j=0; j < N; j++) {  
        if (checkBit(i, j)) {  
            sum += arr[j];  
        }  
    }  
    if (sum == k) {  
        return true;  
    }  
}  
return false;
```

for 3 elements $\rightarrow$ 3 positions $\{0, 1, 2\}$
for N elements $\rightarrow$ N positions $\{0, 1, \dots, N-1\}$

checking if jth bit is set in 'i' or not

e.g. $\begin{matrix} 6 \\ 2 \end{matrix} \rightarrow \begin{matrix} 1 & 1 & 0 \\ & & j \end{matrix}$

T.C = $O(2^N \times N)$ $\rightarrow$ Not good.

S.C = $O(1)$

Advance

Backtracking $\rightarrow$ Dynamic Programming
 $O(2^N)$ $\rightarrow$ $O(N \times k)$

Break :- 10:40 PM

Q3 Given N distinct elements, sum of all subset sums.

e.g. $\{3, 1, 4\}$

$\{ \}$	$\rightarrow$	0
$\{3\}$	$\rightarrow$	3
$\{1\}$	$\rightarrow$	1
$\{4\}$	$\rightarrow$	4
$\{3, 1\}$	$\rightarrow$	4
$\{3, 4\}$	$\rightarrow$	7
$\{1, 4\}$	$\rightarrow$	5
$\{3, 1, 4\}$	$\rightarrow$	8

Ans = 32

idea:- for every subset, iterate and get sum and then add all the sums.

$$T.C = O(2^N * N); S.C = O(1)$$

$\rightarrow$ Contribution Technique.

1) In how many subsets 3 is present? = 4
 $\Rightarrow 3 * 4$

2) In how many subsets 1 is present? = 4
 $\Rightarrow 1 * 4$

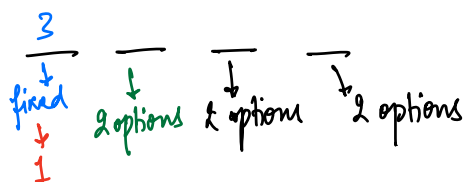
3) In how many subsets 4 is present? = 4
 $\Rightarrow 4 * 4$

$$\text{total} = 12 + 4 + 16 = 32$$

Question to Ask:- In how many subsets, each element is present?

e.g. $\begin{matrix} 0 & 1 & 2 & 3 \\ 3 & 2 & 6 & 8 \end{matrix}$

$\downarrow$
This is present in how many subsets?



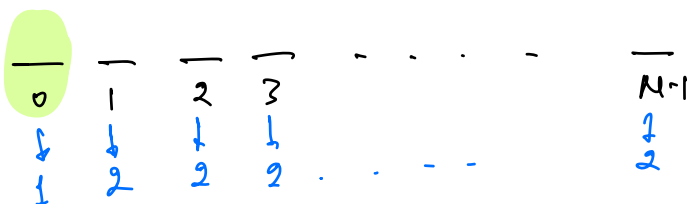
$$\Rightarrow 1 * 2 * 2 * 2 = 8 \text{ subsets will have 3.}$$

- 1 $\{3\}$
- 2 $\{2\}$
- 2 $\{6\}$
- 2 $\{8\}$
- 3 $\{2, 6\}$
- 3 $\{2, 8\}$
- 2 $\{6, 8\}$
- 3 $\{2, 6, 8\}$

In how subsets 2 is present? = 8.

Let's have N elements

In how many subsets 0^{th} index element is present? $= 2^{N-1}$



In how many subsets i^{th} index element is present? $= 2^{N-1}$

int $s = 0$;

for ($i = 0$; $i < N$; $i++$) {

$s = s + arr[i] * (2^{N-1})$

}

return s ;

T.C = $O(N)$

S.C = $O(1)$

Qeta

Given N distinct elements. Calculate $\left(\frac{\text{sum of all subsets}}{\text{sum}} \right) / 2^N$

Constraints:- $1 \leq N \leq 10^5$

$\Rightarrow$ sum of subset sums =

$$= (a_0 * 2^{N-1} + a_1 * 2^{N-1} + a_2 * 2^{N-1} + \dots + a_{N-1} * 2^{N-1}) / 2^N$$

$$= \frac{2^{N-1}}{2^N} (a_0 + a_1 + a_2 + a_3 + \dots + a_{N-1})$$

$$= \frac{(a_0 + a_1 + \dots + a_{N-1})}{2} = \frac{(\text{sum of all array elements})}{2}$$

Ques Given an array, find sum of max. of every subsequence

Arr = 3 1 -4

{ } → 0
 { 3 } → 3
 { 1 } → 1
 { -4 } → -4
 { 3 1 } → 3
 { 3 -4 } → 3
 { 1 -4 } → 1
 { 3 1 -4 } → 3
 Sum = 10

idea:- for every subsequence, get max and add it to your sum.

$$T.C = O(2^N * N) ; S.C = O(1)$$

→ Contribution Technique.

In how many subsequences 3 is maximum = 4
 - - - - - 1 - - - = 2
 - - - - - -4 - - - = 1

$$(3 * 4) + (1 * 2) + (-4 * 1) = 10$$

→ Can we do sorting here? → Yes, we can

{ 3, -4 } OR { -4, 3 }

max/min sum of subsequence does not change

A:- -4 1 3 (sorted array)

In how many subsequences -4 is max

↓ ↓ ↓ = 1 subsequence
 -4 1 1
 ↓ ↓ ↓
 1 Nothing present 1

↓ 1 ↓ ⇒ 2 * 1 * 1 ⇒ 2 subsequences
 2 choices 1 1 { 1 } , { 1, -4 }

$$\begin{array}{c} \underline{\quad} \quad \underline{\quad} \quad \underline{3} \\ \downarrow \quad \downarrow \quad \downarrow \\ 2 \quad 2 \quad 1 \end{array} \Rightarrow 2+2+1 = 4 \text{ subsequences}$$

$$\text{total} \Rightarrow 1*1 + 1*2 + 3*4 = 10 \quad \checkmark$$

Ex. [3 2 6 4 5]

Sort:-

	0	1	2	3	4
	2	3	4	5	6
	↓	↓	↓	↓	↓
	1	2	4	8	16
	↓	↓	↓	↓	↓
	2^0	2^1	2^2	2^3	2^4
	↓	↓	↓	↓	↓
	$2 + 6 + 16 + 40 + 96 = 160$				

Code:-

// Sort in ascending order

```

sum = 0;
for (i = 0; i < N; i++) {
    sum = sum + (arr[i] * (1 <= i));
}
return sum;

```

$$T.C = O(N \log N + N)$$

$$S.C = O(1)$$

ToDo:-

X = sum of max of every subsequence

Y = sum of min of every subsequence → Try this (Very Simple)

$$X - Y = \text{sum of (max - min of every subsequence)}$$

→ Google

$$(max_1 - min_1) + (max_2 - min_2) + (max_3 - min_3)$$

$$(\underline{max_1 + max_2 + max_3}) - (\underline{min_1 + min_2 + min_3})$$